

Table 1: LSDA improvement over baselines for configuration dependency *classification*.

System	cDep	CoT	LLM-RAG
	Mean (improvement)	Mean (improvement)	Mean (improvement)
Precision			
HDFS	0.030 (666.7%)	0.065 (253.8%)	0.133 (72.9%)
HBASE	0.019 (926.3%)	0.049 (298.0%)	0.070 (178.6%)
YARN	0.179 (126.8%)	0.030 (1253.3%)	0.132 (207.6%)
MAPREDUCE	0.047 (461.7%)	0.112 (135.7%)	0.244 (8.2%)
CORE	0.025 (1040.0%)	0.088 (223.9%)	0.147 (93.9%)
CINDER	—	0.038 (802.6%)	0.122 (181.1%)
GLANCE	—	0.140 (10.0%)	0.056 (175.0%)
Average	0.060 (346.7%)	0.075 (257.3%)	0.129 (107.8%)
Recall			
HDFS	0.196 (150.0%)	0.232 (111.2%)	0.188 (160.6%)
HBASE	0.067 (340.3%)	0.097 (204.1%)	0.178 (65.7%)
YARN	0.377 (58.9%)	0.180 (232.8%)	0.280 (113.9%)
MAPREDUCE	0.106 (261.3%)	0.122 (213.9%)	0.204 (87.7%)
CORE	0.239 (100.4%)	0.209 (129.2%)	0.229 (109.2%)
CINDER	—	0.359 (39.8%)	0.235 (113.6%)
GLANCE	—	0.126 (220.6%)	0.083 (386.7%)
Average	0.197 (128.4%)	0.200 (125.0%)	0.200 (125.0%)
F1-score			
HDFS	0.104 (245.2%)	0.078 (360.3%)	0.161 (123.0%)
HBASE	0.057 (682.5%)	0.053 (741.5%)	0.170 (162.4%)
YARN	0.237 (115.2%)	0.081 (529.6%)	0.154 (231.2%)
MAPREDUCE	0.110 (241.8%)	0.091 (313.2%)	0.248 (51.6%)
CORE	0.049 (977.6%)	0.068 (676.5%)	0.213 (147.9%)
CINDER	—	0.044 (1190.9%)	0.141 (302.8%)
GLANCE	—	0.092 (220.7%)	0.114 (158.8%)
Average	0.111 (296.4%)	0.072 (511.1%)	0.165 (166.7%)
AUC			
HDFS	0.670 (12.8%)	0.670 (12.8%)	0.577 (31.0%)
HBASE	0.567 (44.3%)	0.572 (43.0%)	0.632 (29.4%)
YARN	0.674 (26.7%)	0.646 (32.2%)	0.620 (37.7%)
MAPREDUCE	0.599 (32.4%)	0.594 (33.5%)	0.619 (28.1%)
CORE	0.544 (54.2%)	0.610 (37.5%)	0.642 (30.7%)
CINDER	—	0.580 (60.3%)	0.587 (58.4%)
GLANCE	—	0.545 (42.6%)	0.557 (39.5%)
Average	0.620 (32.9%)	0.612 (34.6%)	0.604 (36.4%)